FINAL PROJECT REPORT

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Craigslist Used Car Market Analysis

Data Cleaning, Exploratory Data Analysis, and Interactive Dashboard Development

Course project	Final Project – Unit 3: Data Exploration & Dashboard Development
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Table of Contents

1. Introduction 3
1.1 Project Background 3
1.2 Dataset Overview 3
1.3 Problem Statement 3
1.4 Project Objectives 4
1.5 Scope of the Project 4
2. Dataset Exploration 4
2.1 Data Loading 4
2.2 Initial Data Inspection 4
2.3 Dataset Features 5
2.4 Initial Data Quality Issues 6
3. Data Cleaning and Preprocessing 6
3.1 Feature Selection 7
3.2 Handling Missing Values 7
3.3 Filtering Invalid and Extreme Values 7
3.4 Data Type Conversion 8
3.5 Duplicate Analysis 8
3.6 Outlier Detection 8
3.7 Final Cleaned Dataset 9
4. Exploratory Data Analysis (EDA) 9
4.1 Price Distribution 10
4.2 Vehicle Condition Distribution 10
4.3 Transmission Distribution 10
4.4 Manufacturing Year Distribution 11
4.5 Mileage Distribution 11
4.6 Price vs Manufacturing Year 12
4.7 Price vs Mileage 12
4.8 Average Price by Manufacturer 13
4.9 Correlation Heatmap 14
5. Interactive Streamlit Dashboard 15
5.1 Dashboard Purpose 15
5.2 Tools and Technologies 15
5.3 Dashboard Architecture 15

5.4 Sidebar Filters 16
5.5 Dashboard Components 16
5.6 Dashboard Screenshots 17
5.7 Dashboard Performance 17
6. Key Findings and Discussion 18
6.1 Manufacturing Year and Listing Prices 18
6.2 Mileage and Listing Prices 18
6.3 Differences Between Manufacturers 18
6.4 Vehicle Characteristics and Price Distribution 18
6.5 Outliers and Their Effect on Interpretation 18
6.6 Correlation Versus Causation 18
6.7 Practical Value 19
7. Challenges and Limitations 19
8. Conclusion and Future Work 19
8.1 Conclusion 19
8.2 Future Improvements 20
9. References 20
Appendix 21
A. Additional Chart 21
B. Key Code Snippets 21
C. Dashboard Setup Instructions 22

1. Introduction

1.1 Project Background

The used vehicle market is one of the largest consumer markets in which buyers and sellers set prices with incomplete information. Online classified platforms such as Craigslist publish large numbers of listings in which sellers state an asking price together with descriptive attributes such as model year, mileage, manufacturer and condition. Analysing these listings at scale makes it possible to see how asking prices vary with vehicle characteristics, which attributes are commonly reported, and where the data itself is unreliable.

This project applies a complete data-analysis workflow to a public Craigslist dataset: loading and inspecting the raw data, cleaning it, exploring it statistically and visually, and finally packaging the cleaned data in an interactive Streamlit dashboard so that other users can explore the same questions for any subset of vehicles.

1.2 Dataset Overview

The analysis uses the **Craigslist Cars and Trucks Data** dataset published on Kaggle by Austin Reese [1]. The notebook downloads it with the `kagglehub` library and reads the file `vehicles.csv`. Table 1 summarises the dataset as loaded.

Table 1. *Overview of the source dataset.*

Property	Value
Dataset name	Craigslist Cars and Trucks Data
Source	Kaggle — austinreese/craigslist-carstrucks-data [1]
Original records	426,880 vehicle listings
Original columns	26 (identifiers, listing URLs, vehicle attributes, location, description, posting date)
Main attributes used	price, year, manufacturer, condition, cylinders, fuel, odometer, transmission, type
Purpose	Large real-world sample of used-vehicle listings for price and market-structure analysis

1.3 Problem Statement

Used car asking prices vary widely, from a few hundred dollars to hundreds of thousands, and the raw listings contain missing values, placeholder prices and implausible mileage readings. The analytical problem is therefore twofold: first, to turn the raw listings into a reliable analytical dataset; and second, to identify which vehicle characteristics are associated with higher or lower asking prices. The notebook frames this through five research questions:

1. How does manufacturing year relate to vehicle price?
2. What is the relationship between mileage and price?
3. Which manufacturers have the highest average listing prices?

4. What are the most common vehicle characteristics in the dataset?
5. Are there unusual values or outliers that may affect the analysis?

1.4 Project Objectives

- **Data cleaning:** remove invalid and extreme values, handle missing data and produce a consistent, analysis-ready dataset (`clean_data.csv`).
- **Price analysis:** quantify how listing price relates to model year, mileage and manufacturer using descriptive statistics and Pearson correlation.
- **Visualisation:** communicate distributions and relationships with clearly labelled charts.
- **Dashboard development:** build an interactive Streamlit dashboard that lets users filter the cleaned data and see statistics, charts and insights update.

1.5 Scope of the Project

The project covers descriptive and exploratory analysis of asking prices for the nine selected features, the full cleaning pipeline documented in the notebook, and a dashboard that reproduces and extends the exploratory analysis interactively. It does **not** include predictive modelling or machine learning, geographic analysis (the region, state and coordinate columns were not retained), analysis of the free-text descriptions or images, or any validation of actual transaction prices. Findings describe associations within Craigslist listings and should not be read as causal effects or as statements about the wider used-car market.

2. Dataset Exploration

2.1 Data Loading

The notebook imports Pandas, NumPy, Matplotlib, Seaborn, `pathlib` and SciPy's `stats` module. The dataset is downloaded with `kagglehub.dataset_download("austinreese/craigslist-carstrucks-data")` and the file `vehicles.csv` is read into a DataFrame with `pd.read_csv(file_path, low_memory=False)`; `low_memory=False` lets Pandas infer each column's type from the whole file rather than from chunks, avoiding mixed-type warnings on a file of this size. Displaying the DataFrame confirms **426,880 rows × 26 columns**.

2.2 Initial Data Inspection

Several standard Pandas inspection methods were used before any data was changed:

- **Displaying the DataFrame** (equivalent to `head()`/`tail()`): the first rows contain only `id`, `url`, `region`, `price` and `state`, with almost every vehicle attribute missing, while later rows are fully populated dealer listings. This immediately showed that completeness varies strongly between listings.
- **`shape` / row count:** 426,880 entries indexed 0–426,879 and 26 columns.

- **`info()`:** 2 integer columns (`id`, `price`), 5 float columns (`year`, `odometer`, `county`, `lat`, `long`) and 19 object (text) columns, using about 84.7 MB of memory. `price` is the only vehicle attribute with no missing values.
- **`describe()`:** revealed implausible extremes. Price ranges from $0 to $3,736,928,711 with a mean of $75,199 but a median of only $13,950; odometer ranges from 0 to 10,000,000 miles; model years range from 1900 to 2022.
- **Missing-value analysis:** the percentage of missing values per column was computed with `df.isnull().sum() / len(df) * 100` and sorted (Table 2).

Table 2. *Share of missing values in the raw dataset (selected columns, from the notebook output).*

Column	Missing (%)	Column	Missing (%)
county	100.00	manufacturer	4.13
size	71.77	title_status	1.93
cylinders	41.62	model	1.24
condition	40.79	odometer	1.03
VIN	37.73	fuel	0.71
drive	30.59	transmission	0.60
paint_color	30.50	year	0.28
type	21.75	price	0.00

2.3 Dataset Features

From the 26 raw columns, nine features describing the vehicle and its price were retained for analysis. Table 3 describes them as they appear in the cleaned dataset.

Table 3. *The nine features of the cleaned dataset.*

Feature	Type (cleaned)	Description
price	Integer	Asking price of the listing in US dollars (target variable of the analysis).
year	Integer	Manufacturing (model) year; converted from float to integer during cleaning.
manufacturer	Categorical (text)	Vehicle make, e.g. ford, toyota, ferrari; 42 named makes plus 'unknown'.
condition	Categorical (text)	Seller-reported condition: new, like new, excellent, good, fair, salvage, or 'unknown'.
cylinders	Categorical (text)	Engine cylinders as text (e.g. '6 cylinders'); missing values filled with the mode.
fuel	Categorical (text)	Fuel type: gas, diesel, hybrid, electric, other, or 'unknown'.
odometer	Float	Reported mileage in miles.
transmission	Categorical (text)	automatic, manual, other, or 'unknown'.
type	Categorical (text)	Body type, e.g. sedan, SUV, pickup, truck, coupe, or 'unknown'.

2.4 Initial Data Quality Issues

The inspection identified four main quality problems. First, **missing values** were widespread: `county` was completely empty, and `condition`, `cylinders` and `type` were missing for 22–42% of listings (Table 2). Second, **invalid prices** were common: the minimum price was $0 (typical of placeholder listings) and the maximum exceeded $3.7 billion, which distorts the mean. Third, **unusual mileage** appeared at both ends, with readings of 0 and up to 10,000,000 miles. Fourth, the **price distribution was unreadable** in its raw form: Figure 1 shows the histogram of raw prices, where a handful of extreme values compress almost every listing into a single bar at the left of the axis. No exact duplicate rows were found.

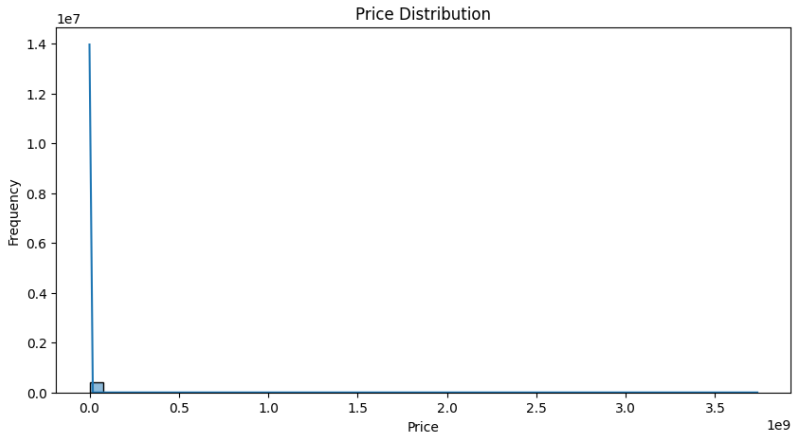


Figure 1. *Raw price distribution before cleaning. Extreme values (up to $3.7 billion) compress almost all listings into the first bin.*

3. Data Cleaning and Preprocessing

The cleaning steps were executed in the order shown in Table 4. The subsections below discuss each type of operation; where the required report structure groups steps differently, the notebook order is noted.

Table 4. *Cleaning pipeline in notebook order.*

Step	Notebook operation	Records after step
1	Drop the empty `county` column	426,880
2	Check for exact duplicate rows (none found)	426,880
3	Keep prices between $1,000 and $200,000	—
4	Keep odometer ≤ 500,000 miles (missing kept for now)	—
5	Remove rows with |z| ≥ 3 on log(1 + price)	377,943 (48,937 removed in steps 3–5)
6	IQR outlier analysis on price (outliers identified, retained)	377,943
7	Select the nine analysis features	377,943
8	Drop rows missing price, year or odometer	374,762 (3,181 removed)
9	Fill missing categories with 'unknown'; cylinders with the mode	374,762
10	Duplicate checks on IDs and selected features; convert year to integer	374,762
11	Validation checks and export to `clean_data.csv`	374,762

3.1 Feature Selection

After the value filters, the notebook keeps nine columns: `price`, `year`, `manufacturer`, `condition`, `cylinders`, `fuel`, `odometer`, `transmission` and `type`. These directly describe the vehicle and its price and are relevant to the research questions. Identifiers and links (`id`, `url`, `region_url`, `image_url`, `VIN`), free text (`description`, `model`) and location fields (`region`, `state`, `lat`, `long`, `posting_date`) were excluded because they do not describe the vehicle in a way that can be compared across listings without further processing. `size` was excluded because 71.8% of its values were missing.

3.2 Handling Missing Values

Different strategies were applied depending on the role of each feature:

- **Essential numeric fields** — rows missing `price`, `year` or `odometer` were removed with `dropna(subset=[...])`, because these values cannot be meaningfully imputed for a price analysis. This removed 3,181 rows (377,943 → 374,762).
- **Categorical fields** — missing `manufacturer`, `condition`, `fuel`, `transmission` and `type` were filled with the explicit label `'unknown'`, on the reasoning that a missing value most likely means the seller did not specify it. This keeps the listings while making the gap visible in later analysis.
- **Cylinders** — missing values were filled with the most frequent category (the mode), treating cylinders as a discrete feature.

After these steps, `df_clean.isnull().sum()` reports zero missing values in all nine columns.

3.3 Filtering Invalid and Extreme Values

Three filters were applied to the full dataset before feature selection:

1. **Price range:** `df[(df['price'] >= 1000) & (df['price'] <= 200_000)]` removes placeholder and implausibly low prices as well as extreme high values.
2. **Mileage limit:** `df[df['odometer'].isna() | (df['odometer'] <= 500_000)]` removes impossible odometer readings while temporarily keeping missing values, which are handled later.
3. **Log-price z-score:** `np.abs(stats.zscore(np.log1p(df['price']))) < 3` removes the remaining extremes measured on a logarithmic scale, which suits right-skewed price data.

Together these filters reduced the data from 426,880 to **377,943** listings (48,937 removed). The notebook notes that the bounds are analytical assumptions that may exclude some legitimate listings. Figure 2 shows the price distribution after filtering, which is now readable and clearly right-skewed.

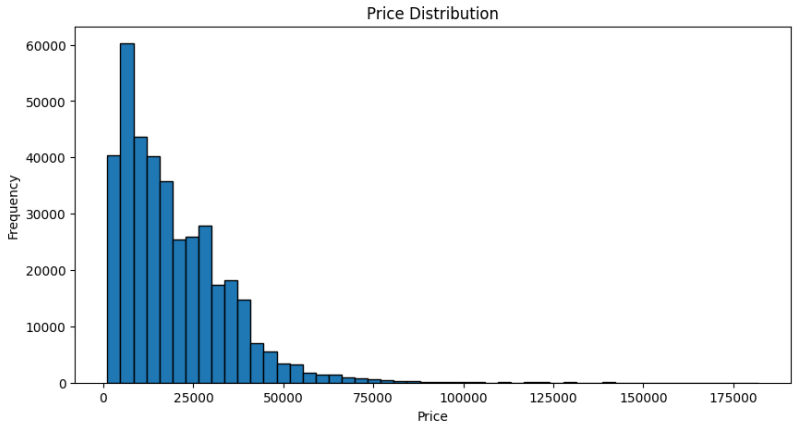


Figure 2. *Price distribution after price, mileage and z-score filtering (377,943 listings).*

3.4 Data Type Conversion

The `year` column was stored as a float in the raw file because of missing values. After rows with missing years were removed, it was converted to an integer with `df_clean['year'].astype(int)`. The final types are integer for `price` and `year`, float for `odometer`, and text (object) for the six categorical features.

3.5 Duplicate Analysis

Duplicates were examined at three levels. A full-row check on the raw data (`df.duplicated().sum()`) found **0** duplicates, and the listing identifier `id` had **0** duplicate values. However, **159,494** rows of the nine-feature dataset share exactly the same combination of values with another row. These were deliberately retained: once identifiers, descriptions and locations are removed, two different listings of, for example, a 2017 Ford pickup in good condition at the same price and mileage band can legitimately produce identical rows. Removing them would discard real listings and bias the distributions towards rarer vehicles.

3.6 Outlier Detection

Two complementary outlier methods were used:

- **Z-score (removal):** applied to `log1p(price)` with a threshold of $|z| < 3$ as part of the filtering step in Section 3.3. Rows outside this band were removed.

- **Interquartile range (identification only):** on the filtered data, Q1 = $7,995 and Q3 = $27,990 (IQR = $19,995), giving bounds of −$21,997.5 and $57,982.5. The notebook identified **7,821** price outliers above the upper bound; the box plot in Figure 3 shows them as a long tail of points.

The IQR outliers were **retained**, because prices above $58,000 are plausible for luxury, collectible and specialty vehicles. The notebook acknowledges that these values influence averages, which is why both mean and median are reported in the analysis.

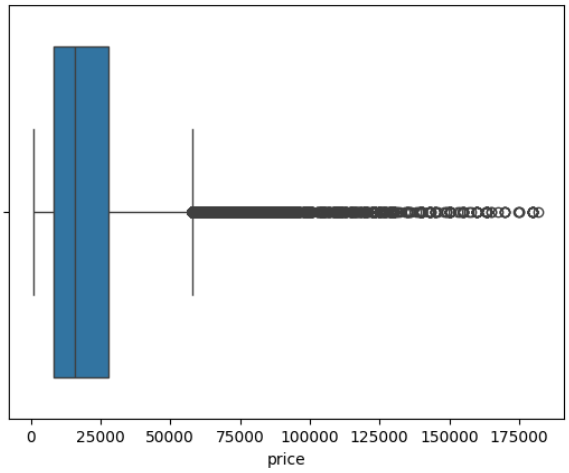


Figure 3. *Box plot of listing prices after filtering; points to the right of the whisker are IQR outliers (retained).*

3.7 Final Cleaned Dataset

The final validation cell confirms 0 non-positive prices, 0 years outside 1900–2026, 0 negative odometer readings and 0 missing values. Table 5 summarises the result.

Table 5. *Cleaned dataset summary (verified against the notebook outputs and clean_data.csv).*

Metric	Value
Original records	426,880
Records after value filtering	377,943
Final records	374,762
Removed records	52,118 (12.2%)
Final features	9
Remaining missing values	0

The cleaned DataFrame was exported with `df_clean.to_csv("clean_data.csv", index=False)`, which writes the nine columns without the Pandas index. This file is the single input to the dashboard.

4. Exploratory Data Analysis (EDA)

This section discusses the remaining notebook charts (Figures 4–11), together with the filtered price distribution already shown in Figure 2. The distribution charts in Sections 4.2–4.5 were created on the

377,943 filtered listings, before rows with missing year or odometer were removed and before missing categories were labelled 'unknown'; the scatter plots, manufacturer comparison and heatmap use the final 374,762-record dataset. Counts quoted in the text are computed from the final cleaned dataset.

4.1 Price Distribution

After cleaning, listing prices range from $1,125 to $182,000. The mean is **$19,522** and the median **$15,995**, and the middle 50% of listings lie between **$7,995** and **$27,990**. Figure 2 shows a right-skewed distribution: most listings are concentrated below about $30,000, with a long, thinning tail of expensive vehicles. Because the mean is pulled upwards by this tail, the median is the better summary of a typical asking price.

4.2 Vehicle Condition Distribution

Figure 4 shows the reported conditions. Among listings that state a condition, **good** is the most frequent (117,511 listings in the cleaned data, 31.4% of all records), followed by **excellent** (90,436; 24.1%) and **like new** (19,232; 5.1%). Fair, new and salvage are rare. Condition is unreported for **140,589** listings (37.5%), which are labelled 'unknown' in the cleaned data and do not appear in this chart because it was drawn before imputation. The dominance of 'good' and 'excellent' suggests that sellers tend to describe their vehicles favourably.

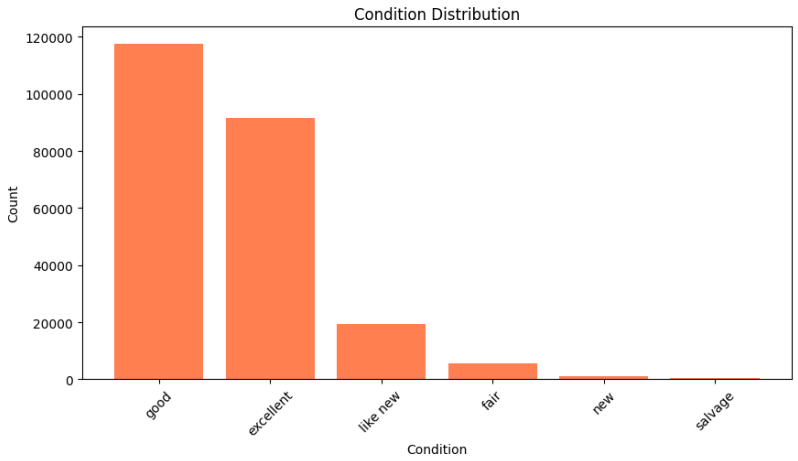


Figure 4. *Distribution of reported vehicle condition (listings with a stated condition).*

4.3 Transmission Distribution

As Figure 5 shows, automatic transmissions dominate the market: 290,478 listings (77.5%) in the cleaned data. The 'other' category is the second largest (60,501; 16.1%), larger than manual (22,274; 5.9%). The size of 'other' indicates that many sellers or dealer feeds did not use the standard categories, which limits how precisely transmission can be analysed.

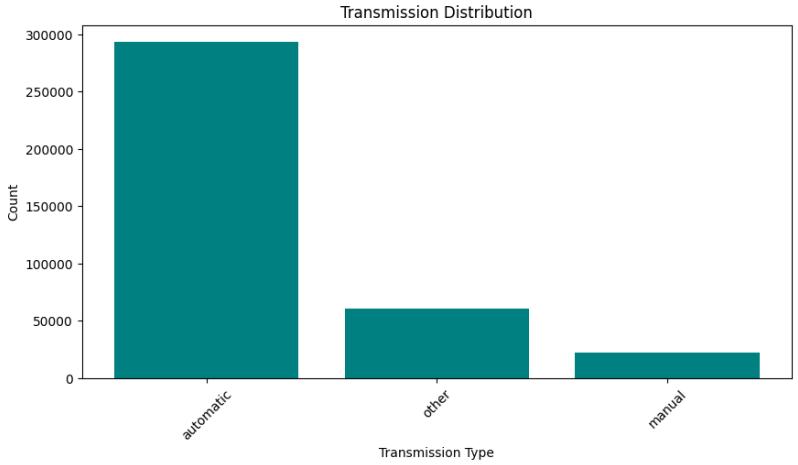


Figure 5. *Distribution of transmission types.*

4.4 Manufacturing Year Distribution

Figure 6 shows that listings are heavily concentrated in recent model years. In the cleaned data the median model year is 2013, the middle 50% span 2008–2017, and 93.9% of vehicles were built in 2000 or later. A thin tail extends back to 1900, consisting of classic and collectible vehicles; these older listings are few but, as Section 4.6 shows, often carry high prices.

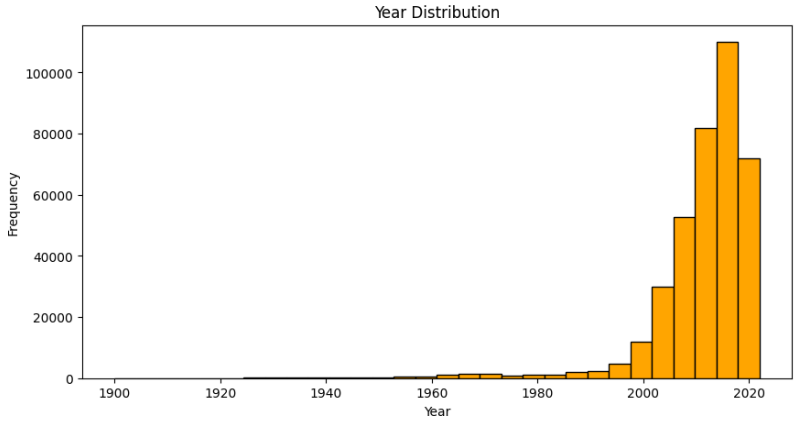


Figure 6. *Distribution of manufacturing years.*

4.5 Mileage Distribution

Figure 7 shows the odometer distribution. Mileage is right-skewed with a median of **87,488 miles** and a mean of 92,854 miles; the middle 50% of listings lie between 38,351 and 135,500 miles, and 94.6% have 200,000 miles or fewer. The tallest bars are at very low mileage, which reflects near-new dealer stock but also includes 928 listings with an odometer reading of exactly 0, some of which are likely data-entry placeholders.

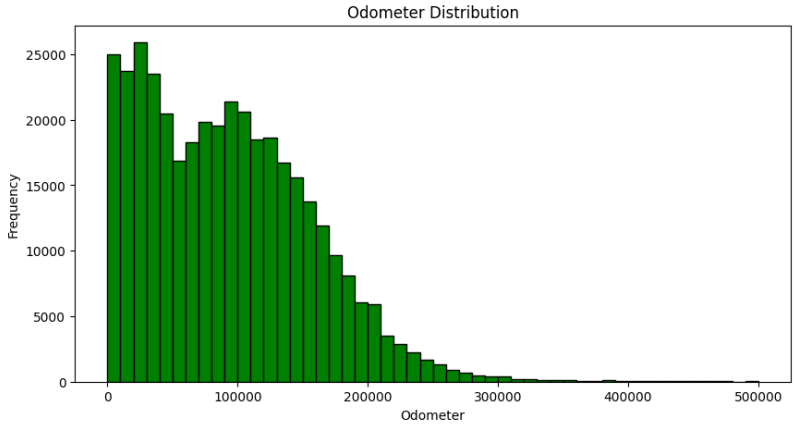


Figure 7. *Distribution of odometer readings (miles) after the 500,000-mile limit.*

4.6 Price vs Manufacturing Year

Figure 8 plots price against model year for a random sample of 10,000 listings (`random_state=42`). The Pearson correlation computed on all 374,762 listings is **r = +0.353**, which was re-verified from `clean_data.csv` for this report. This is a moderate positive relationship: newer vehicles tend to be listed at higher prices. Splitting the cleaned data by model-year quartiles makes the difference concrete: listings from 2017 or newer have a median price of $29,991, compared with $6,495 for model year 2008 or older. The relationship is far from perfect, because prices vary widely within each year and because classic vehicles from before 1980 frequently command high prices, which weakens a straight-line relationship.

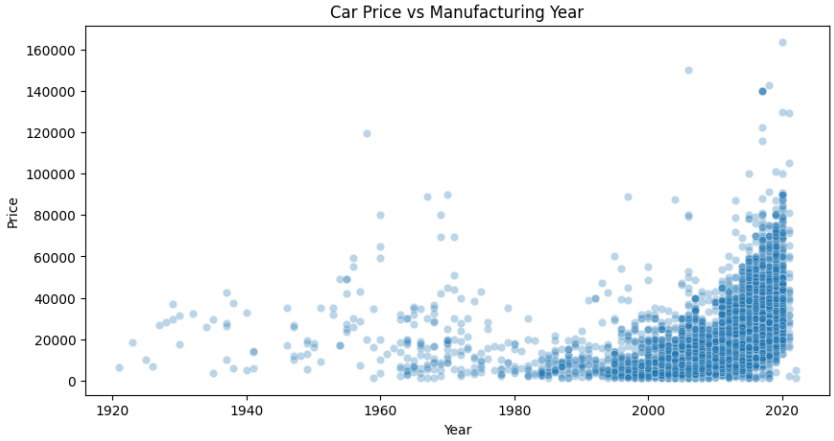


Figure 8. *Listing price versus manufacturing year (random sample of 10,000 listings).*

4.7 Price vs Mileage

Figure 9 plots price against odometer reading for the same sampling approach. The correlation on the full cleaned dataset is **r = −0.517** (re-verified), a moderate negative relationship and the strongest linear relationship with price among the numeric features. The lowest-mileage quarter of listings (≤ 38,351 miles) has a median price of $29,990, while the highest-mileage quarter (≥ 135,500 miles) has a median of $6,995. The chart also shows a dense band of low-price listings across all mileages, indicating that mileage alone does not determine price.

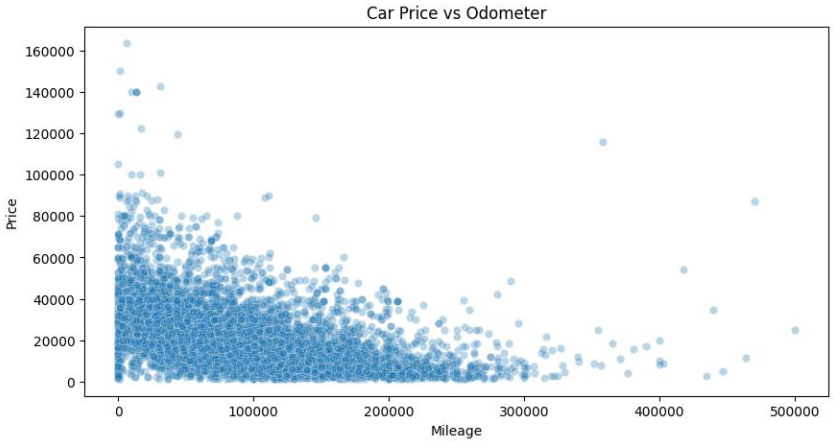


Figure 9. *Listing price versus mileage (random sample of 10,000 listings).*

4.8 Average Price by Manufacturer

Figure 10 ranks the ten manufacturers with the highest mean listing price in the cleaned data; Table 6 adds the median and the number of listings behind each average, which the chart does not show.

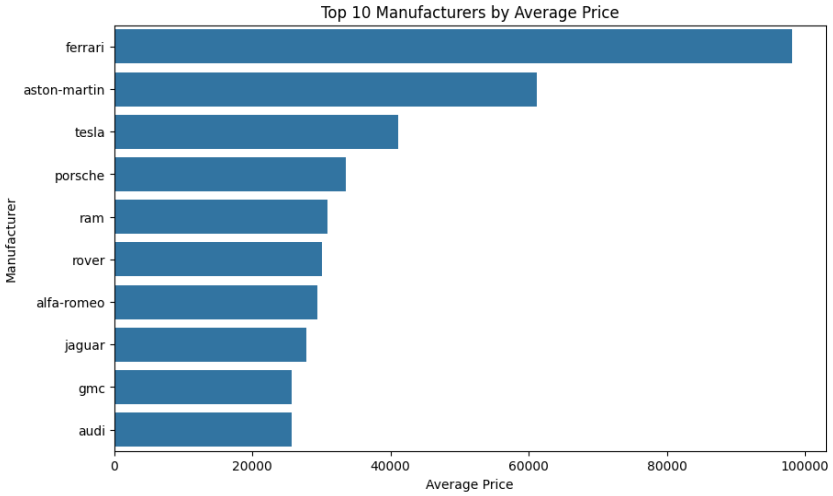


Figure 10. *Top 10 manufacturers by average listing price.*

Table 6. *Top 10 manufacturers by average price, with median and listing count (cleaned dataset).*

Rank	Manufacturer	Average price	Median price	Listings
1	Ferrari	$98,108	$98,900	64
2	Aston Martin	$61,137	$52,000	21
3	Tesla	$41,124	$37,990	809
4	Porsche	$33,510	$29,997	1,253
5	RAM	$30,919	$30,499	15,820
6	Rover	$30,141	$26,995	1,887
7	Alfa Romeo	$29,482	$29,590	852
8	Jaguar	$27,796	$28,990	1,858
9	GMC	$25,648	$24,590	14,935
10	Audi	$25,638	$24,990	6,934

Ferrari, Aston Martin and Tesla lead the ranking, consistent with their positioning as luxury or premium brands. However, the averages for Ferrari (64 listings) and Aston Martin (21 listings) rest on very small samples and are sensitive to individual listings, whereas RAM and GMC appear in the top ten with roughly 15,000 listings each, reflecting the strong prices of trucks. For comparison, the most-listed manufacturers are Ford (62,619 listings), Chevrolet (48,439) and Toyota (30,357); Figure 15 in the Appendix shows the ten most frequent manufacturers.

4.9 Correlation Heatmap

Figure 11 shows the Pearson correlation matrix for the three numeric features of the cleaned dataset, and Table 7 lists the values, all re-verified from `clean_data.csv`. A positive coefficient means two variables tend to increase together; a negative coefficient means one tends to decrease as the other increases; values near zero indicate no linear relationship.

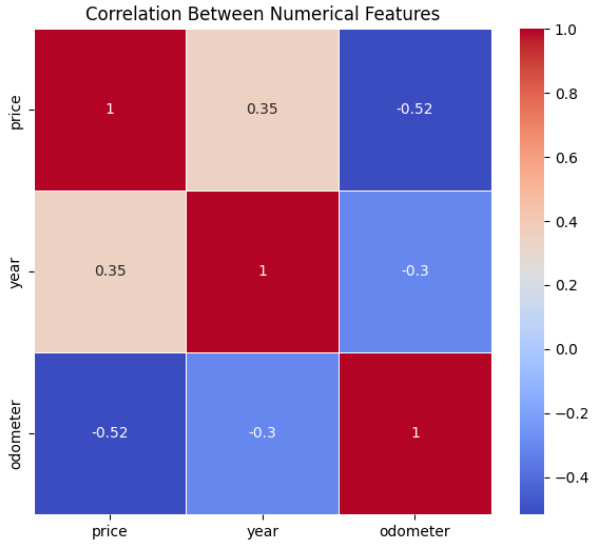


Figure 11. *Correlation heatmap of price, manufacturing year and odometer.*

Table 7. *Pairwise Pearson correlations (cleaned dataset).*

Variable pair	Pearson r	Interpretation
Price – Manufacturing year	+0.353	Moderate positive: newer vehicles tend to be listed higher.
Price – Odometer	−0.517	Moderate negative: higher mileage tends to mean a lower price.
Manufacturing year – Odometer	−0.304	Moderate negative: older vehicles have usually accumulated more miles.

The negative correlation between model year and odometer means that age and mileage partly overlap as explanations of price: some of the price difference attributed to mileage is also related to vehicle age, and vice versa. A simple correlation matrix cannot separate these effects.

5. Interactive Streamlit Dashboard

5.1 Dashboard Purpose

The notebook answers the research questions for the dataset as a whole. The dashboard extends this by letting a user ask the same questions of any subset, for example only Toyota pickups, only vehicles under $10,000, or only diesel vehicles from 2015 onwards, and see every statistic, chart and written insight recalculate. It also makes the analysis accessible to people who do not run Python notebooks.

5.2 Tools and Technologies

Table 8. *Technologies used in app.py.*

Technology	Role in the application
Streamlit	Web application framework: page layout, sidebar widgets, tabs, tables, caching (`st.cache_data`) and the CSV download button [7].
Pandas	Loading `clean_data.csv`, filtering with boolean masks, aggregations (group-by, value counts, describe) and correlation [2].
NumPy	Vectorised filtering masks, server-side histogram binning (`np.histogram`), percentiles and linear trend fitting (`np.polyfit`) [3].
Plotly	Interactive charts with hover tooltips: Plotly Express for scatter plots and Graph Objects for bars, donuts, histogram and heatmap [6].

5.3 Dashboard Architecture

The application (`app.py`) follows a load–filter–render structure:

1. **Load:** `load_data()` reads `clean_data.csv` once and caches it with `@st.cache_data`. It checks that the file exists, can be parsed, and contains all nine expected columns, and shows a clear error message otherwise. Text columns are stored as Pandas categoricals for faster filtering; values are not changed.

2. **Filter:** the sidebar builds a dictionary of filter values, and `apply_filters()` combines them into a single vectorised boolean mask. The cached source data is never modified; an empty multiselect means "all".
3. **Render:** the filtered DataFrame is passed to section functions that compute statistics and build Plotly figures. All KPIs, correlations, rankings and insights are calculated from the full filtered data. If no listings match, the dashboard shows a warning instead of empty charts.

5.4 Sidebar Filters

Table 9. *Sidebar filters implemented in app.py.*

Filter	Control	Behaviour
Manufacturer	Multiselect	Any combination of the 43 manufacturer values; empty = all.
Manufacturing year	Range slider	1900–2022 in steps of one year.
Price (USD)	Range slider	$1,000–$182,000 in $500 steps (bounds rounded outward so no record is excluded by default).
Mileage (odometer)	Range slider	0–500,000 miles in 1,000-mile steps.
Vehicle condition	Multiselect	Ordered from new to salvage, plus 'unknown'.
Fuel type	Multiselect	gas, diesel, hybrid, electric, other, unknown.
Transmission	Multiselect	automatic, manual, other, unknown.
Vehicle type	Multiselect	All body types in the data.

The sidebar also contains a short description of the dataset, a **Reset filters** button that restores every filter to its default, a live count of matching listings with their share of the 374,762 records, and a link to the Kaggle source.

5.5 Dashboard Components

- **KPI cards:** total listings, average price, median price, average mileage and average model year for the selection, each with the full-dataset value shown for reference.
- **Overview tab:** a data preview (first ten filtered rows) and summary statistics for the numeric features (count, mean, standard deviation, minimum, median, maximum) and categorical features (number of values, most frequent value and its share).
- **Price Distribution:** histogram with an adjustable number of bins (10–200), optional log-scaled counts, and average and median reference lines.
- **Price Drivers:** scatter plots of price against manufacturing year and against mileage, coloured by grouped condition or by the three most-listed manufacturers, with a linear trend line and the Pearson correlation for the current selection alongside the full-dataset value.
- **Manufacturers:** horizontal bar chart of the top manufacturers ranked by average price, median price or listing count, with an adjustable number shown, a minimum-listings threshold, and listing counts in the hover tooltip.
- **Vehicle Characteristics:** condition and body-type bar charts and fuel and transmission donut charts.

- **Correlation:** heatmap of price, year and odometer with written interpretation; correlations that cannot be computed (for example, when a single model year is selected) are shown as 'n/a'.
- **Key Insights & Findings:** sentences generated from the filtered data about age, mileage, manufacturers, condition, price distribution and the typical listing, plus a separate panel with the full-dataset findings.
- **Data Explorer:** text search across the categorical columns (and model year), sorting, a choice of how many rows to display, and a **CSV download** of all matching records.

5.6 Dashboard Screenshots

Screenshots of the running dashboard were not available when this report was prepared. The placeholders below indicate where they should be inserted; each can be replaced in Word by selecting the box and inserting the image.

[SCREENSHOT PLACEHOLDER]

Insert a screenshot of the full dashboard with the default (unfiltered) view.

Figure 12. *Dashboard overview: sidebar filters, header and KPI cards (placeholder).*

[SCREENSHOT PLACEHOLDER]

Insert a screenshot showing a scatter plot with its correlation value.

Figure 13. *Price Drivers tab: price vs manufacturing year and price vs mileage (placeholder).*

[SCREENSHOT PLACEHOLDER]

Insert a screenshot demonstrating how results change when filters are applied.

Figure 14. *Manufacturers or Key Insights tab with filters applied (placeholder).*

5.7 Dashboard Performance

Because the dataset has about 375,000 rows, the application includes several optimisations:

- **Caching:** the CSV is read once with `@st.cache_data` instead of on every interaction; the CSV export is also cached for repeated downloads of the same selection.
- **Vectorised filtering:** all filters are combined into one NumPy boolean mask, and categorical dtypes make `isin` comparisons fast.
- **Sampling:** scatter plots draw a reproducible random sample of 10,000 points by default (`random_state=42`, adjustable from 1,000 to 30,000), while correlations and trend lines use every filtered row; the chart caption states this.

- **Server-side aggregation:** the histogram is binned with `np.histogram`, so only the bars, not every price, are sent to the browser.

6. Key Findings and Discussion

6.1 Manufacturing Year and Listing Prices

Model year has a moderate positive association with asking price (r = +0.353). Newer vehicles are typically listed for considerably more: the median price of vehicles from 2017 onwards is about 4.6 times that of vehicles from 2008 or earlier. The relationship is weakened by classic vehicles, whose prices rise rather than fall with age.

6.2 Mileage and Listing Prices

Mileage shows the strongest linear relationship with price among the numeric features (r = −0.517). Low-mileage listings have a median price more than four times that of high-mileage listings. Because mileage and model year are themselves correlated (r = −0.304), part of this relationship reflects vehicle age.

6.3 Differences Between Manufacturers

Average prices differ substantially by manufacturer. Exotic and premium brands (Ferrari, Aston Martin, Tesla, Porsche) have the highest averages, while truck-oriented brands (RAM, GMC) rank highly with very large samples. Mainstream manufacturers dominate the number of listings, with Ford, Chevrolet and Toyota together accounting for over a third of the cleaned data. Rankings based on small samples should be interpreted cautiously.

6.4 Vehicle Characteristics and Price Distribution

The typical listing is a gas-powered (83.8%) vehicle with an automatic transmission (77.5%), most often described as being in good or excellent condition and built after 2008. Prices are right-skewed: the mean ($19,522) exceeds the median ($15,995), so averages overstate the price of a typical vehicle. Substantial shares of listings do not report condition (37.5%) or body type (21.2%).

6.5 Outliers and Their Effect on Interpretation

The raw data contained extreme prices up to $3.7 billion and mileage up to 10 million, which made raw summaries meaningless. After filtering, 7,821 high-price IQR outliers were kept as plausible luxury and specialty listings. They raise the mean price and the averages of small manufacturer groups, which is why medians are reported alongside means throughout this report and the dashboard.

6.6 Correlation Versus Causation

All relationships reported here are correlations within Craigslist listings. They show that newer and lower-mileage vehicles tend to be listed at higher prices, but they do not prove that age or mileage cause those prices: brand, body type, condition, region and seller type all vary together with these

features. Pearson correlation also captures only linear relationships, so non-linear patterns such as the price premium of classic vehicles are understated.

6.7 Practical Value

For buyers and sellers, the findings provide reference points: typical asking prices by model year, mileage and manufacturer, and the knowledge that mileage is at least as informative as age. For analysts, the project shows how much cleaning crowd-sourced listing data requires before it can be summarised, and the dashboard provides a reusable tool for exploring specific market segments.

7. Challenges and Limitations

- **Asking versus selling prices:** Craigslist prices are what sellers ask, not what buyers pay; negotiated transaction prices are not available.
- **Missing information:** condition (40.8%), cylinders (41.6%) and type (21.8%) were frequently missing in the raw data. Labelling them 'unknown' preserves the rows but limits analysis by these features, and filling cylinders with the mode inflates the '6 cylinders' category (63.4% of the cleaned data).
- **Selection bias from filtering:** the $1,000–$200,000 price range, the 500,000-mile limit and the z-score filter are analytical choices that removed 48,937 listings, some of which may have been legitimate (for example, inexpensive project cars).
- **Unequal manufacturer sample sizes:** manufacturer averages range from 21 listings (Aston Martin) to 62,619 (Ford), so averages for small groups are unstable.
- **Data-entry quality:** user-entered values include 928 listings with zero mileage, very old model years and broad 'other' categories for transmission and fuel.
- **Unavailable details:** vehicle history, trim level, accident records and model-level specifications are not part of the cleaned dataset, and location fields were not analysed.
- **Limits of correlation:** Pearson correlation measures only linear association between pairs of variables and cannot separate the overlapping effects of age and mileage.
- **Duplicates:** 159,494 rows share identical feature values; they were kept because listing IDs are unique, but some could represent re-posted listings.

On the technical side, the main challenge was the size of the dataset in an interactive application. Re-reading the CSV or plotting every point on each interaction would make the dashboard slow, which was addressed through the caching, sampling and aggregation described in Section 5.7.

8. Conclusion and Future Work

8.1 Conclusion

This project analysed 426,880 Craigslist vehicle listings. A documented cleaning pipeline removed invalid and extreme values, handled missing data and produced a dataset of 374,762 records with nine features and no missing values, exported as `clean_data.csv`. Exploratory analysis showed that prices

are right-skewed (median $15,995), that newer vehicles are moderately associated with higher prices (r = +0.353), that higher mileage is moderately associated with lower prices (r = −0.517), and that premium manufacturers such as Ferrari, Aston Martin and Tesla have the highest average prices, although on small samples. An interactive Streamlit dashboard makes these analyses available for any subset of the data through eight sidebar filters, KPI cards, interactive Plotly charts, automatically generated insights and a searchable data explorer with CSV export.

8.2 Future Improvements

The following extensions are proposed; none of them is implemented in the current project:

- **Geographic market analysis:** reintroduce the region, state and coordinate fields to compare prices across locations and map regional differences.
- **Additional vehicle characteristics:** include model, drive type, paint colour and title status, and parse model names to compare specific vehicles.
- **More advanced interactive filtering:** linked chart selections (for example, clicking a manufacturer to filter all charts) and saved filter presets.
- **Predictive price modelling:** train and evaluate regression models to estimate a fair price from vehicle attributes, with proper validation.
- **Improved data quality handling:** detect placeholder odometer values and re-posted listings using descriptions or VINs.

9. References

[1] A. Reese, "Craigslist Cars and Trucks Data," Kaggle. [Online]. Available: https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data

[2] The pandas development team, "pandas documentation." [Online]. Available: https://pandas.pydata.org/docs/

[3] NumPy Developers, "NumPy documentation." [Online]. Available: https://numpy.org/doc/

[4] Matplotlib Development Team, "Matplotlib documentation." [Online]. Available: https://matplotlib.org/stable/

[5] M. Waskom, "seaborn: statistical data visualization — documentation." [Online]. Available: https://seaborn.pydata.org/

[6] Plotly Technologies Inc., "Plotly Python graphing library documentation." [Online]. Available: https://plotly.com/python/

[7] Snowflake Inc., "Streamlit documentation." [Online]. Available: https://docs.streamlit.io/

[8] SciPy Developers, "scipy.stats.zscore — SciPy documentation." [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.zscore.html

[9] Kaggle, "kagglehub." GitHub repository. [Online]. Available: https://github.com/Kaggle/kagglehub

Appendix

A. Additional Chart

Figure 15 shows the ten manufacturers with the most listings after value filtering, complementing the price ranking in Section 4.8.

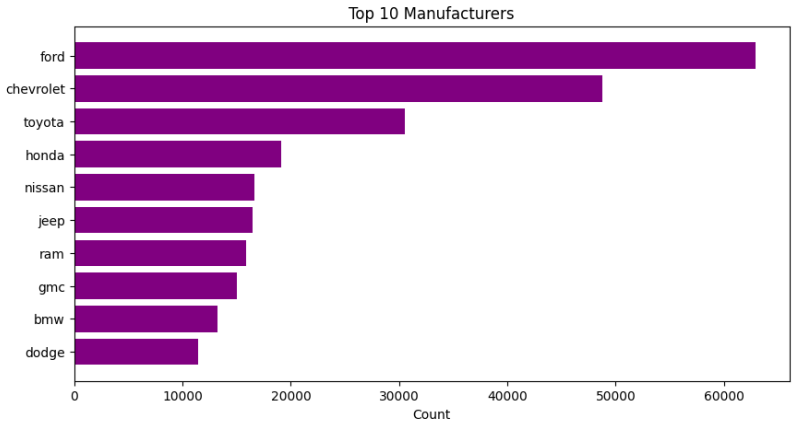


Figure 15. *Top 10 manufacturers by number of listings.*

B. Key Code Snippets

Notebook — value filtering and z-score outlier removal:

```
df = df[(df['price'] >= 1000) & (df['price'] <= 200_000)]
df = df[df['odometer'].isna() | (df['odometer'] <= 500_000)]
z = np.abs(stats.zscore(np.log1p(df['price'])))
df = df[z < 3].reset_index(drop=True)
```

Notebook — feature selection, missing values and export:

```
cols = ['price', 'year', 'manufacturer', 'condition', 'cylinders',
        'fuel', 'odometer', 'transmission', 'type']
df_clean = df[cols].copy()
df_clean = df_clean.dropna(subset=['price', 'year', 'odometer'])
for col in ['manufacturer', 'condition', 'fuel', 'transmission', 'type']:
    df_clean[col] = df_clean[col].fillna('unknown')
df_clean['cylinders'] = df_clean['cylinders'].fillna(df_clean['cylinders'].mode()[0])
df_clean['year'] = df_clean['year'].astype(int)
df_clean.to_csv("clean_data.csv", index=False)
```

Dashboard — cached loading and vectorised filtering (simplified from app.py):

```
@st.cache_data(show_spinner="Loading the cleaned Craigslist dataset…")
def load_data(path: str) -> pd.DataFrame:
    ...  # existence and column checks, numeric coercion, categorical dtypes

def apply_filters(df, f):
    mask = np.ones(len(df), dtype=bool)
    for col in ("manufacturer", "condition", "fuel", "transmission", "type"):
        if f.get(col):
            mask &= df[col].isin(f[col]).to_numpy()
    for col in ("year", "price", "odometer"):
        lo, hi = f[col]
        mask &= (df[col].to_numpy() >= lo) & (df[col].to_numpy() <= hi)
    return df.loc[mask]
```

C. Dashboard Setup Instructions

The project folder contains `app.py`, `clean_data.csv`, `requirements.txt` (Streamlit 1.40–1.49, Pandas 2.x, NumPy, Plotly 5.x), `README.md`, a theme file `.streamlit/config.toml` and the notebook. On Windows, from the project folder:

```
py -m pip install -r requirements.txt
py -m streamlit run app.py
```

Streamlit then opens the dashboard at http://localhost:8501. On macOS or Linux, use `python3` in place of `py`. `clean_data.csv` must be in the same folder as `app.py`.